AI Prompts

AI tools used: Claude , Antigravity , MetaAi

Prompt 1: Domain Modeling and Data Dictionary

Act as a Principal Data Architect in pharmaceutical cold-chain logistics. We are building "ColdChain Sentinel", a compliance and deviation management platform for FDA 21 CFR Part 11, `USP <1079>`, and EU GDP.

Define the data model for 12 entities: Product, Lot, Packaging, Logger, Shipment, ShipmentLot, RouteStop, CustodyEvent, TemperatureReading, ThresholdPolicy, ExcursionReport, GroundTruth.

Use these ID formats: `LOT-2026-000001`, `SHP-2026-000001`, `LOG-<shipment-id>-01`, `TEMP-<logger-id>-<timestamp>`, `EXCR-2026-000001`.

Provide:

- a) `coldchain/models.py` with typed dataclasses and Enums,
- b) `data_dictionary.md` listing type, required/optional, allowed values, units, uniqueness and validation rules for every field.

Prompt 2: Compliance Engine

Act as a Computational Biophysicist and Regulatory QA Specialist. Write `coldchain/compliance.py` with:

- Mean Kinetic Temperature per `USP <1079>` using ΔH/R = 10,000 K (83.144 kJ/mol). Do not use a linear average.
- Formula: `T_MKT = 10000 / (-ln((1/n) * sum(exp(-10000 / (T_i + 273.15))))) - 273.15`.
- Deviation checks: missing logger (CRITICAL, SENSOR_UNASSIGNED); product band conflicts with policy (CRITICAL, THRESHOLD_MISMATCH); reading gap over 30 minutes (MODERATE, DATA_TRANSMISSION_GAP); unverified signature or broken seal (CUSTODY_INTEGRITY_BREACH); foreign logger or shipment IDs in telemetry (IOT_CROSS_SHIPMENT_LEAKAGE).
- Disposition, applied in this priority order when several rules fire: freeze at or below 0°C for products stored at 2-8°C (REJECT_AND_DESTROY), then cumulative excursion over the maximum (QUARANTINE_FOR_ASSAY), then MKT over the limit (RELEASE_WITH_CAUTION).
- A 64-character SHA-256 evidence hash of all readings and custody events.

Prompt 3: Synthetic Data Generator

Act as a Senior Backend Systems Engineer. Write a deterministic Python script (`coldchain/generator.py`) that:

- Takes a seed (default 20261022) and a scale: starter = 50 shipments, full = 5,000 shipments and 1,000 loggers.
- Generates parents first (Product, Packaging, Policy), then Lot, Logger, Shipment and Stops, then Custody and Readings, then ExcursionReport.
- Creates readings every 10 minutes over 24 to 60 hours per shipment, as a random walk that reverts toward the container target, with spikes during airport tarmac dwells and a ±0.15°C sensor offset.
- Injects five labelled edge cases into a separate `ground_truth.csv` and `.json`, never into the input features: logger not assigned, threshold mismatch (mRNA at -80°C under a 2-8°C policy), telemetry blackout over 30 minutes, custody gap, and cross-shipment contamination (5 foreign readings).
- Computes durations, peak, low and MKT using `compliance.py`.

Prompt 4: Database and Tests

Build the persistence layer and tests:

- `coldchain/db.py`: SQLite schema with `PRAGMA foreign_keys = ON`, composite indexes on (shipment_id, timestamp), (logger_id) and (primary_lot_id), and export of all 12 tables to `exports/csv/` and `exports/json/`.
- `tests/test_coldchain.py` with three tests: constant temperature gives identical MKT while fluctuating temperature gives MKT above the arithmetic mean; an unassigned logger is flagged CRITICAL; the end-to-end validator passes with zero defects.

Prompt 5: Validation and Reproducibility

Implement the audit engine:

- `coldchain/validator.py` with 8 checks: primary key uniqueness, foreign key integrity, allowed enum values, date order (manufacture < departure < arrival < expiry), arithmetic reconciliation (peak, low, cumulative minutes, MKT), SHA-256 evidence hashes, duplicate rows, and precision, recall and F1 against `ground_truth`.
- `coldchain/verify_reproducibility.py`: generate the data twice with seed 20261022 and assert the SHA-256 table digests match exactly.
- `cli.py` with subcommands `generate`, `validate` and `verify`.

Prompt 6: Streamlit Dashboard

Act as a Lead Frontend Engineer. Build `app.py` with Streamlit and Altair.

- Sidebar persona switcher: Quality Assurance Lead (batch quarantines, e-signature sign-off, downloadable Certificate of Conformance); Logistics Analyst (telemetry rates, ΔH/R sensitivity slider, ±0.15°C uncertainty bands); Carrier / Warehouse Operator (route dwells, digital handover

sign-off); Planner (lane risk matrix, packaging alerts); Supplier (lot inventory, batch approval status).

- Five tabs: Command Center (KPI banner, severity breakdown, shipment ledger); Thermal Telemetry (time series with limits, MKT line and uncertainty band); Chain of Custody (handovers with signature badges, tarmac dwell chart); CAPA Studio (root cause, CAPA plan, one-click certificate); Benchmark (confusion matrix and recall scorecard).

Prompt 7: Documentation

Generate the documentation pack:

- `database_UML.md`: Mermaid ERD, UML class diagram and constraints matrix.
- `validation_summary.md`: audit report covering all 8 checks and the reproducibility checksums.
- `architecture.md`: block diagrams, data flow, and two sequence diagrams (generation and compliance evaluation).
- `subject_guide.md`: guide to the biopharma cold chain, `USP <1079>`, 21 CFR Part 11, Arrhenius physics and thermal packaging.
- `source_manifest.md`: file inventory, seed documentation and SHA-256 table digests.